

Reasoning Language Models

Introduction

Contents

1. **What is reasoning?**

- a. Reasoning and Decomposition

2. **Prompting Methods, Inference-Time reasoning**

- a. Chain of Thought prompting
- b. Train of Thoughts
- c. Analogical Prompting

3. **Training for reasoning**

- a. RLHF Recap
- b. Rewards and Bootstrapping, STaR, ORM, PRM
- c. Reinforcement Learning with Verifiable Rewards
- d. DeepSeek-R1?

4. **Test-time compute?**

5. **Conclusion and critical questions**

What is reasoning?

Reasoning and Decomposition

Program Induction by Rationale Generation: Learning to Solve and Explain Algebraic Word Problems. [1]

Problem 1:

Question: Two trains running in opposite directions cross a man standing on the platform in 27 seconds and 17 seconds respectively and they cross each other in 23 seconds. The ratio of their speeds is:

Options: A) $\frac{3}{7}$ B) $\frac{3}{2}$ C) $\frac{3}{88}$ D) $\frac{3}{8}$ E) $\frac{2}{2}$

Correct Option: B

Reasoning and Decomposition

Problem 1:

Question: Two trains running in opposite directions cross a man standing on the platform in 27 seconds and 17 seconds respectively and they cross each other in 23 seconds. The ratio of their speeds is:

Options: A) $3/7$ B) $3/2$ C) $3/88$ D) $3/8$ E) $2/2$

Rationale: Let the speeds of the two trains be x m/sec and y m/sec respectively. Then, length of the first train = $27x$ meters, and length of the second train = $17y$ meters. $(27x + 17y) / (x + y) = 23 \rightarrow 27x + 17y = 23x + 23y \rightarrow 4x = 6y \rightarrow x/y = 3/2$.

Correct Option: B

Figure 1: Example of the reasoning problems in *Program Induction by Rationale Generation: Learning to Solve and Explain Algebraic Word Problems*. [1]. This paper, published at ACL in 2017, is the first one to use natural language to describe intermediate reasoning steps.

Reasoning and Decomposition

Thinking, Fast and Slow. [2]

1. **System 1 reasoning:** fast, automatic and unconscious; e.g. recognizing faces, taking quick decisions
2. **System 2 reasoning:** slow, deliberate, analytical; e.g. solving maths problems

System 1 is prone to **cognitive biases**, System 2 is **rational**. How do machines pass **from System 1 to System 2**?

Reasoning and Decomposition

Prompting strategies: inference time reasoning

Chain of Thought Prompting

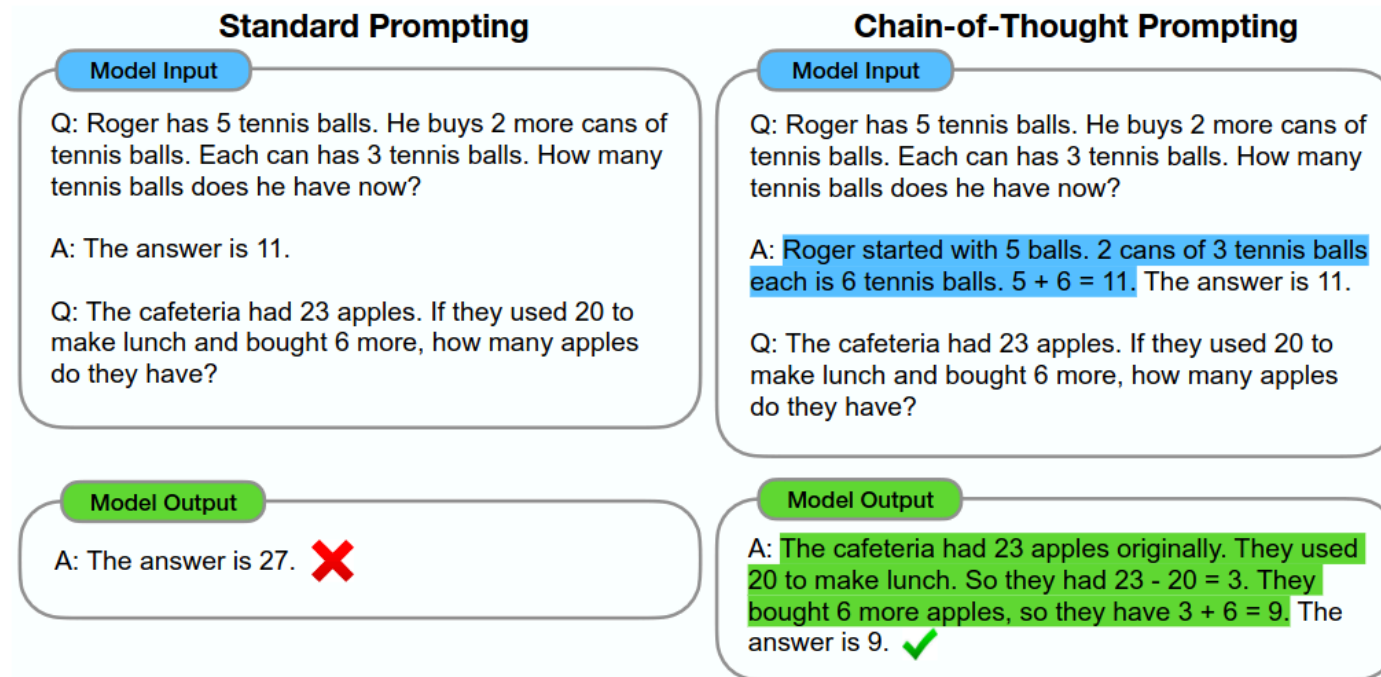


Figure 2 : The 2021 NeurIPS paper by Wei et al. introduced *Chain of Thought (CoT) prompting*, which enables large language models to tackle complex arithmetic, commonsense, and symbolic reasoning tasks. Chain-of-thought reasoning processes are highlighted.

Chain of Thought prompting

(a) Few-shot

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The answer is 8. ✗

(b) Few-shot-CoT

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are $16 / 2 = 8$ golf balls. Half of the golf balls are blue. So there are $8 / 2 = 4$ blue golf balls. The answer is 4. ✓

(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: The answer (arabic numerals) is

(Output) 8 ✗

(d) Zero-shot-CoT (Ours)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: **Let's think step by step.**

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

Example inputs and outputs of GPT-3. [3]

Chain of Thought prompting [4]

CoT introduces intermediate reasoning steps $z_1, \dots, z_n$ between input x and output y :

$$x \rightarrow z_1 \rightarrow \dots \rightarrow z_n \rightarrow y$$

Each thought is generated sequentially:

$$z_i \sim p_{\theta}^{CoT}(z_i \mid x, z_{1:i-1})$$

The final answer is generated conditioned on the full chain:

$$y \sim p_{\theta}^{CoT}(y \mid x, z_{1:n})$$

Chain of Thought prompting [4]

Summary: reasoning steps and the answer are sampled as a single continuous sequence:

$$[z_1, \dots, z_n, y] \sim p_{\theta}^{CoT}(z_1, \dots, z_n, y \mid x)$$

The **decomposition** of thoughts (phrase, sentence, paragraph, ...) is left ambiguous.

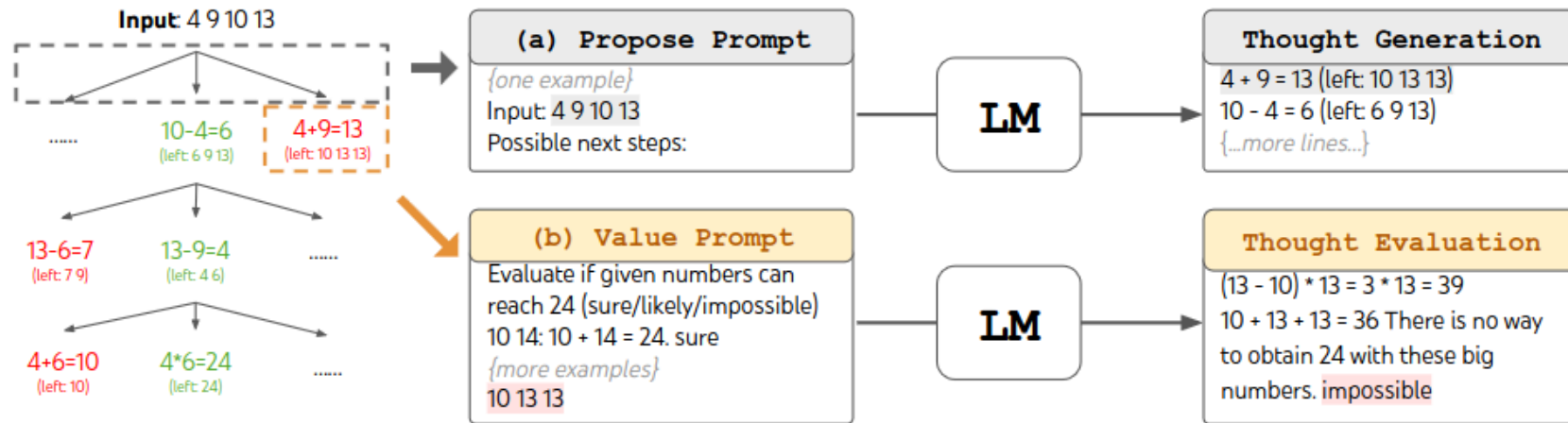
Tree of Thoughts

Deliberate Problem Solving with Large Language Models [4]

1. How to decompose the intermediate process into thought steps
2. How to generate potential thoughts from each state
3. How to heuristically evaluate states
4. What search algorithm to use

Tree of Thoughts

Deliberate Problem Solving with Large Language Models [4]



Example of ToT in a game of 24. The LM is prompted for (a) thought generation and (b) valuation.

Analogical Prompting

Large Language Models as Analogical Reasoners [5]

0-shot

Model Input

Q: What is the area of the square with the four vertices at $(-2, 2)$, $(2, -2)$, $(-2, -6)$, and $(-6, -2)$?

0-shot CoT

Model Input

Q: What is the area of the square with the four vertices at $(-2, 2)$, $(2, -2)$, $(-2, -6)$, and $(-6, -2)$?

Think step by step.

- Generic guidance of reasoning

Few-shot CoT

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have?

A: Roger started with 5 balls. 2 cans of 3 balls each is 6 balls. The answer is $5 + 6 = 11$.

...

Q: What is the area of the square with the four vertices at $(-2, 2)$, $(2, -2)$, $(-2, -6)$, and $(-6, -2)$?

- Need labeled exemplars of reasoning

Analogical Prompting (Ours)

Model Input

Q: What is the area of the square with the four vertices at $(-2, 2)$, $(2, -2)$, $(-2, -6)$, and $(-6, -2)$?

Instruction:

Recall relevant exemplars:

Solve the initial problem:

Model Output

Relevant exemplars:

Q: What is the area of the square with a side length of 5?

A: The area of a square is found by squaring the length of its side. So, the area of this square is $5^2 = 25$

Solve the initial problem:

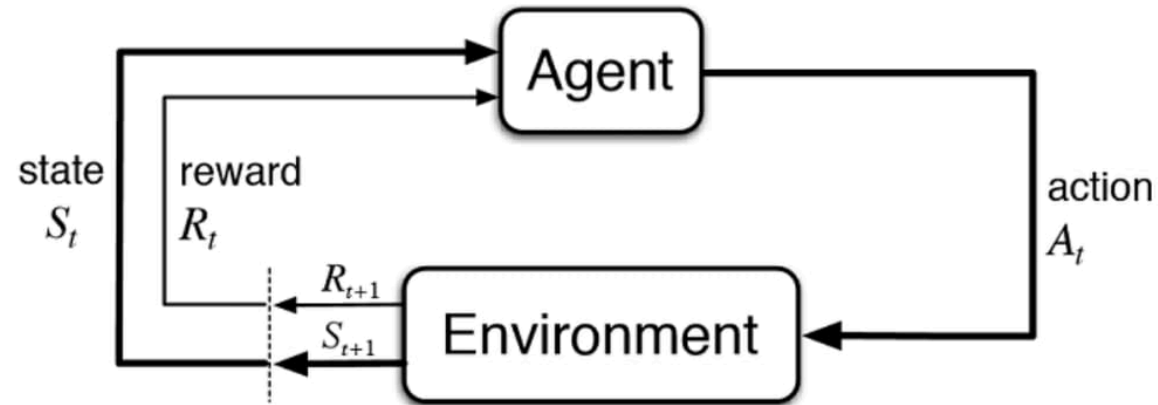
To find the area of the square, we need to find the side length. The length is $\dots \sqrt{(2 - (-2))^2 + (-2 - 2)^2} = \sqrt{32}$. So, the area of the square is $(\sqrt{32})^2 = 32$.

- Exemplars are automatically generated and tailored to each problem



Reinforcement Learning Recap

Reinforcement Learning Recap



In reinforcement learning, an agent learns from the environment by interacting with it through trial and error and receiving rewards as feedback for performing actions.

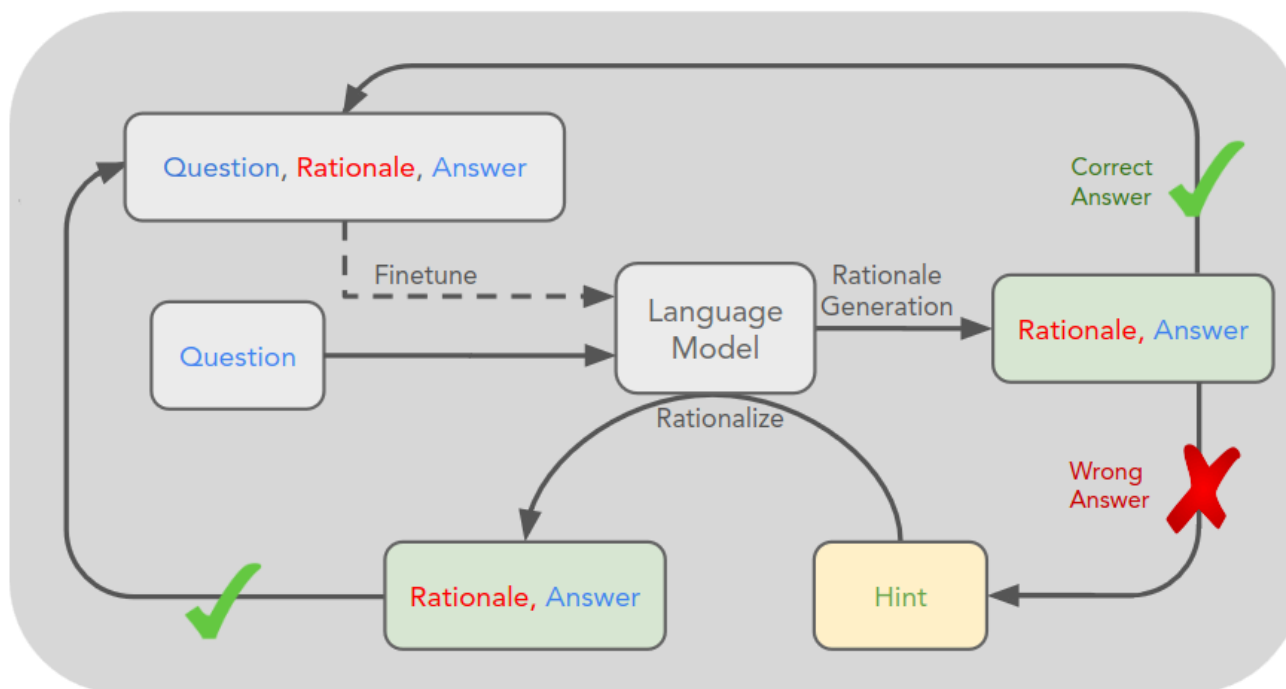
The agent does not know the environment beforehand (black box): this is why it is a **learning** process.

Reinforcement Learning

- **Environment**, with which the agent interacts, outputs **observations**
- Agent makes **decisions**
- Agent gets new **observations**
- Agent gets a **rewards** based on how good (or bad) the actions were.

Training for Reasoning

STaR: Bootstrapping Reasoning with Reasoning [6]



Q: What can be used to carry a small dog?
Answer Choices:
(a) swimming pool
(b) basket
(c) dog show
(d) backyard
(e) own home
A: The answer must be something that can be used to carry a small dog. Baskets are designed to hold things. Therefore, the answer is basket (b).

An overview of STaR and a STaR-generated rationale on CommonsenseQA. Fine-tuning outer loop is the dashed line. Questions and ground truth answers in the dataset, rationales generated using STaR. [4]

STaR: Bootstrapping Reasoning With Reasoning [6]

Input

- Pretrained LLM **M**
- Dataset:

$$D = \{(x_i, y_i)\}$$

- Few-shot rationale examples:

$$P = \{(x_i^p, r_i^p, y_i^p)\}$$

STaR: Bootstrapping Reasoning With Reasoning [6]

Iterative loop

1. Generate rationale:

$$M(x_i, P) \rightarrow (\hat{r}_i, \hat{y}_i)$$

2. Filter:

$$\hat{y}_i = y_i$$

3. Fine-tune on:

$$(x_i, \hat{r}_i, y_i)$$

4. Repeat

RL in STaR

The LLM first generates a rationale r and then predicts an answer y .

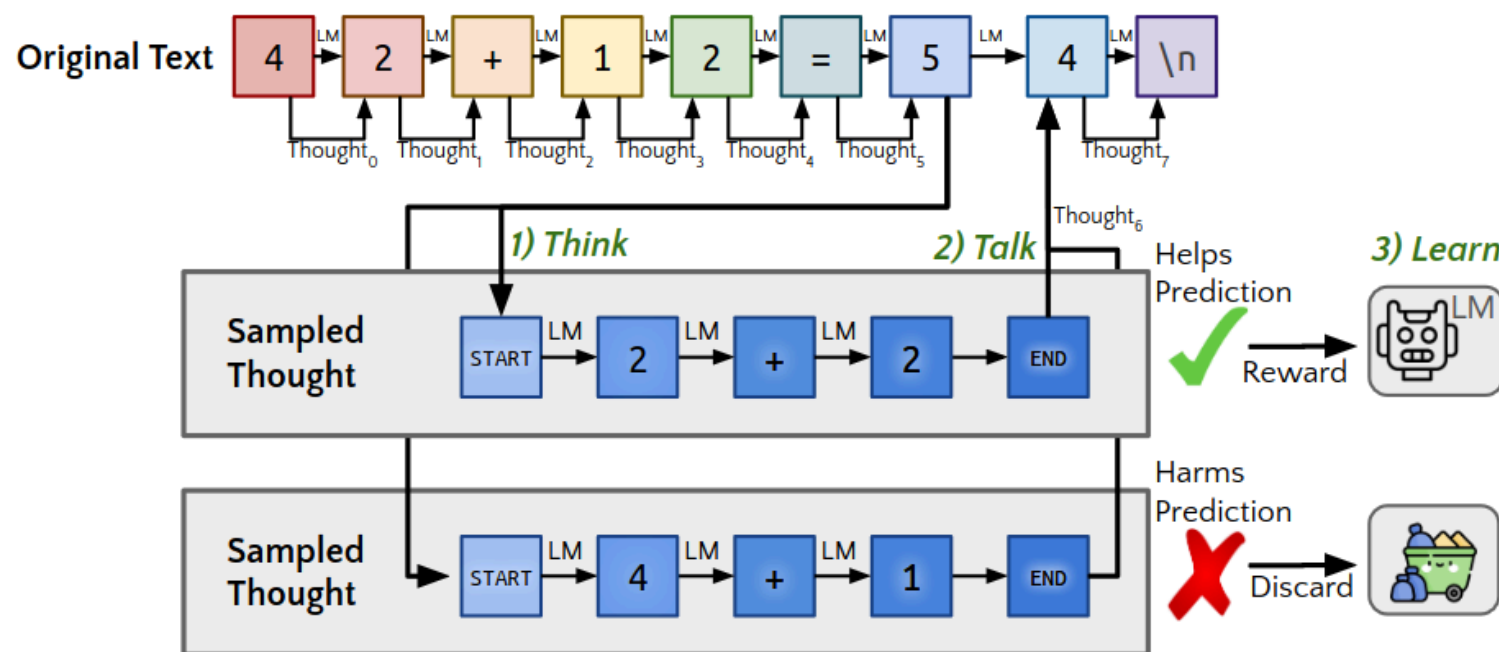
$$p_M(y|x) = \sum_r p(r|x)p(y|x, r)$$

STaR approximates maximizing:

$$J(M) = \sum_i \mathbb{E}_{\hat{r}_i, \hat{y}_i} [\mathbb{1}(\hat{y}_i = y_i)]$$

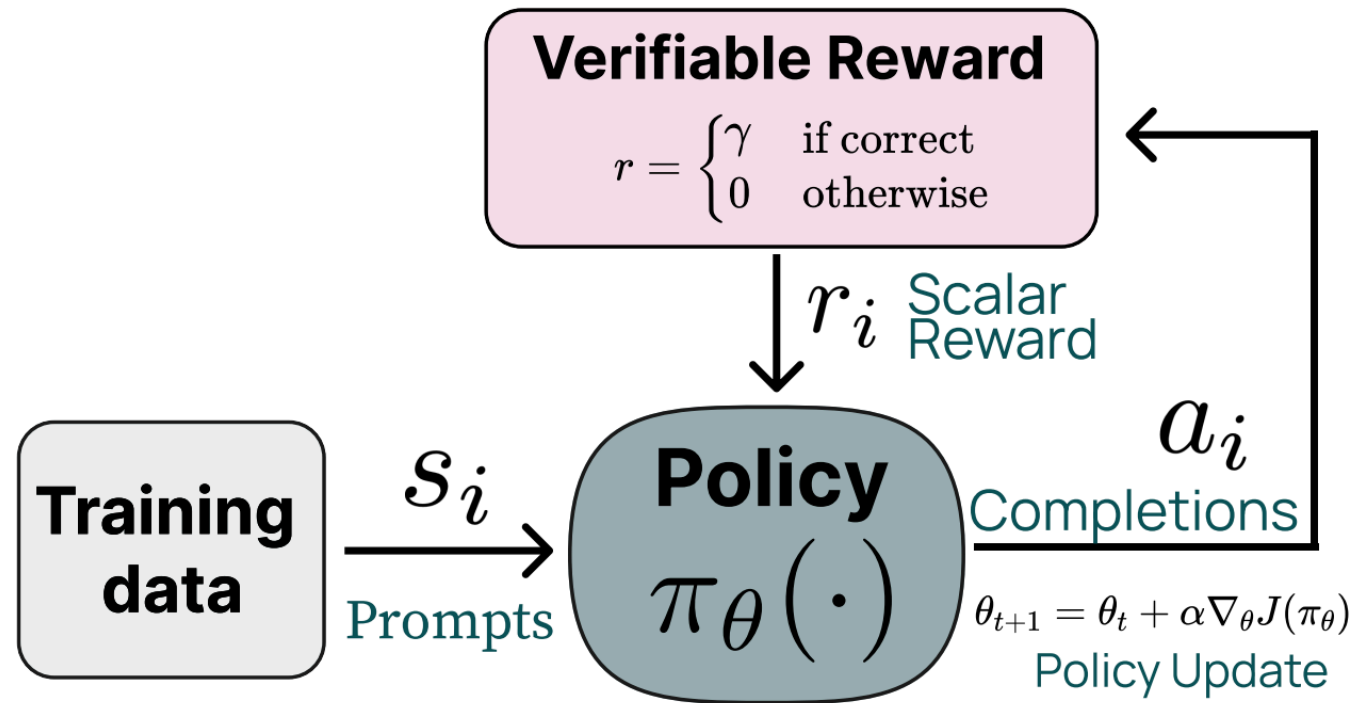
where the reward is **1 only for correct answers**.

Quiet-STaR: Language Models Can Teach Themselves to Think Before Speaking [5]



Reinforcement Learning with Verifiable Rewards (RLVR)

Tulu 3: Pushing Frontiers in Open Language Model Post-Training. [8]



RLVR

The model is trained on tasks for which the correctness can be (easily) verified, either in an exact, mathematical way (formal proofs) or through executables (e.g. unit tests).

There is no uniform way of defining a task as verifiable; for example, mathematical proofs are (intuitively) among the most robustly verifiable tasks.

RLVR: Verifying Function

Given a question x , a generated solution trajectory is:

$$\tau = (r_1, \dots, r_T, y)$$

where $r_{1:T}$ denotes the reasoning trace and y the final answer.

The verifier V_ϕ assigns a score to a candidate solution:

$$V_\phi(x, \tau) \rightarrow s \in [0, 1]$$

RLVR: Verifying Function

Training data consists of solution trajectories with correctness labels:

$$\mathcal{D}_V = \{(x_i, \tau_i, c_i)\}_{i=1}^N, \quad c_i \in \{0, 1\}$$

The verifier is optimized

Questions?

References -- TODO: Correct Bibliography

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[7] Wen et al. Reinforcement Learning with Verifiable Rewards Implicitly Incentivizes Correct Reasoning in Base LLMs.

[8] Lambert et al. Tülu 3: Pushing Frontiers in Open Language Model Post-Training. 2025